

```

void handleData() {
  DynamicJsonDocument doc(1024);
  doc["motion"] = digitalRead(RCWL_PIN);
  doc["lux"] = map(analogRead(LIGHT_SENSOR_PIN), 0, 4095, 0, 1000);
  doc["light"] = light_on;
  doc["bright"] = brightness;
  doc["autoRadar"] = auto_radar;
  doc["autoLight"] = auto_light;
  doc["time"] = wifi_connected ? current_time : "Connecting...";
  doc["temp"] = wifi_connected ? current_temp : "--°C";
  doc["weather"] = wifi_connected ? current_weather : "Unavailable";
  doc["aqi"] = wifi_connected ? current_aqi : "AQI: --";

  String out;
  serializeJson(doc, out);
  server.send(200, "application/json", out);
}

```

Fig. 5: Data-handle function

automatically synchronises and retrieves real-time data, such as time, AQI (air quality index), and weather conditions, when connected to Wi-Fi with internet access. Fig. 3 illustrates the structure of the web dashboard UI.

Through these functionalities, the device also operates as an IoT dashboard or smart clock, providing remote control access, live time, AQI readings, and weather updates through a single interface.

In the code, the Wi-Fi SSID for the required network is configured first. The mode is then configured to operate the device as AP or STA. The current implementation enables both, although this consumes more current and power. A function is created to check the radar sensor data in real time and update it on the cards. The light sensor integrated inside the coin board is also monitored.

The UI supports manual or automatic light control, with options for ambient light or radar input. Fig.

4 shows the code snippet used to configure AP and STA, SSID, and password. Fig. 5 shows the data-handling function.

Construction and testing

After uploading the source code to the development board, assembly of the radar-based SSR (solid-state relay) unit can begin as per the schematic. The RCWL-0516 micro-wave radar module is mounted securely, and its output pin is connected either directly to the controller's IO pin or through the optocoupler path using the solder-jumper selection.

The VO2223 solid-state relay IC is then placed, followed by connection of the series resistor and snubber capacitor, as shown in Fig. 2, to ensure smooth AC load triggering. The 2-terminal block for the bulb or AC appliance is attached, guaranteeing correct routing of the Live (L) and Neutral (N) wires. If included in the design, the FPC camera interface is connected

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through the dedicated connector to provide additional sensing capability.

Once wiring is completed, the board is powered through the VIN terminal, and initial testing is performed to validate detection, triggering, and load-switching functions.

After powering the device, connect the bulb to the AC Load Terminal block, then connect the device to Wi-Fi. On a phone, the user can open Wi-Fi settings, locate the device network, and connect to it using the SSID and password configured earlier.

A browser is then opened, and the device IP address is entered. The web UI appears, providing configuration and monitoring options. The system can function as a human-motion alert device, with radar data displayed in the first card and light-sensor data in the next. The toggle button allows manual light control. If automatic switching based on radar data is required, select the radar auto mode. If automatic switching based on ambient light is preferred, select the auto-on light option. **EFY**

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